Impedance and Admittance Control of a 2-DOF Robot Arm

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I. Introduction & Motivation

In this project, we study this problem on the 2-DOF robot arm shown in Figure 1. Its equations of motion contain nonlinear terms (sines and cosines) and make it a simple but representative manipulator.

Our primary goal is to design and analyze a torque-based *impedance controller* [1], which makes the arm behave like a virtual spring-damper about a desired trajectory. We will (i) derive and linearize the open-loop dynamics, (ii) show how the impedance law changes the local dynamics and stability, and (iii) evaluate the controller in simulation under reference changes and external disturbance torques. Throughout, we carry one concrete example arm so that every matrix below has numbers we can check.

II. System, Inputs & Notation

A. Example arm

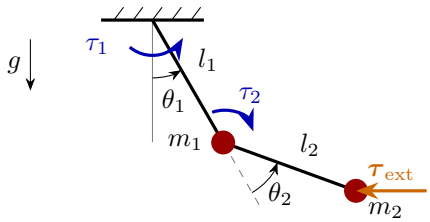


Fig. 1. 2-DOF arm (adapted from [2]). Joint angles θ_1, θ_2 are zero when the arm hangs straight down; θ_2 is relative to link 1. The controller commands the motor torques τ_1 (shoulder) and τ_2 (elbow) in blue. The environment acts through the external torque $\boldsymbol{\tau}_{\text{ext}}$ (orange), which we cannot command.

For all numerical examples we use point masses at the link tips with $m_1 = m_2 = 1$ kg, $l_1 = l_2 = 1$ m, and $g = 9.81$ m/s^2.

B. How we interact with the arm

The arm has two actuators, one at each joint. Our *control input* is the vector of motor torques

$$\mathbf{u} = [\tau_1 \;\; \tau_2]^T \in \mathbb{R}^2,$$

where τ_1 acts at the shoulder (on link 1) and τ_2 at the elbow (between link 1 and link 2). Because each joint has its own motor, both degrees of freedom are directly actuated. The environment (contact, or a push on the arm) acts through the *external torque* $\boldsymbol{\tau}_{\text{ext}} \in \mathbb{R}^2$. It is a disturbance: we do not choose it, and it is zero in free motion. We assume we measure the joint angles and velocities, so the controller sees the full state. In the impedance controller of Section IV, the only other signal we supply is the desired joint configuration $\boldsymbol{\theta}_d$, which sets the equilibrium the virtual spring pulls toward.

C. Notation

Table I lists every symbol used in this proposal. Vectors and matrices are bold; a dot is a time derivative.

TABLE I
Symbols and example values.

Symbol	Meaning	Example
θ_1, θ_2	joint angles (shoulder, elbow); $\theta = 0$ is hanging down	rad
$\dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}}$	joint velocity, acceleration	rad/s, rad/s^2
$m_i,\ l_i$	mass and length of link i	1 kg, 1 m
g	gravitational acceleration	9.81 m/s^2
$\mathbf{M}(\boldsymbol{\theta})$	inertia matrix	2×2
$\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$	Coriolis/centrifugal matrix	2×2
$\boldsymbol{\tau}_g(\boldsymbol{\theta})$	gravity torque vector	N·m
$\mathbf{u} = [\tau_1 \;\; \tau_2]^T$	motor torques (control input)	N·m
$\mathbf{B}$	maps $\mathbf{u}$ to joint torques; $\mathbf{I}_2$ here	2×2
$\boldsymbol{\tau}_{\text{ext}}$	external (contact) torque, uncommanded	N·m
$\mathbf{x}$	state $[\boldsymbol{\theta}^T \;\; \dot{\boldsymbol{\theta}}^T]^T$	$\mathbb{R}^4$
$\mathbf{f}(\mathbf{x}, \mathbf{u})$	state-space dynamics, $\dot{\mathbf{x}} = \mathbf{f}$	
$(\mathbf{x}^*, \mathbf{u}^*)$	equilibrium state–input pair	$(\mathbf{0}, \mathbf{0})$
$\mathbf{A}, \mathbf{B}_{\text{lin}}$	Jacobians of $\mathbf{f}$ at $(\mathbf{x}^*, \mathbf{u}^*)$	4×4, 4×2
$\boldsymbol{\theta}_d,\ \mathbf{e}$	desired angles; error $\boldsymbol{\theta} - \boldsymbol{\theta}_d$	rad
$\mathbf{K}, \mathbf{D}$	virtual stiffness and damping gains	2×2

D. Control objective

Given a constant desired configuration $\boldsymbol{\theta}_d$, design a state-feedback law $\mathbf{u} = \boldsymbol{\pi}(\mathbf{x}, \boldsymbol{\theta}_d)$ such that (i) with $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$ the error $\mathbf{e} = \boldsymbol{\theta} - \boldsymbol{\theta}_d$ converges to zero, and (ii) when $\boldsymbol{\tau}_{\text{ext}} \neq \mathbf{0}$ the arm yields like a spring-damper, $\mathbf{K}\mathbf{e} = \boldsymbol{\tau}_{\text{ext}}$ at rest, with $\mathbf{K}$ and $\mathbf{D}$ chosen by us.

III. Mathematical Model & Linearization

A. Dynamics

As given by [2], the equations of motion (EOM) for a general manipulator with joint angles $\boldsymbol{\theta} \in \mathbb{R}^n$ are

$$\boxed{\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} = \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} + \boldsymbol{\tau}_{\text{ext}}.} \qquad (1)$$

Compared with [2], the term $\boldsymbol{\tau}_{\text{ext}}$ is added so that interaction with the environment can be studied. For our arm ($n = 2$), with $c_2 = \cos\theta_2$, $s_1 = \sin\theta_1$, $s_2 = \sin\theta_2$, $s_{12} = \sin(\theta_1 + \theta_2)$,

$$\mathbf{M}(\boldsymbol{\theta}) = \begin{bmatrix} (m_1 + m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 c_2 & m_2 l_2^2 + m_2 l_1 l_2 c_2 \\ m_2 l_2^2 + m_2 l_1 l_2 c_2 & m_2 l_2^2 \end{bmatrix}$$

$$\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) = \begin{bmatrix} 0 & -m_2 l_1 l_2 (2\dot{\theta}_1 + \dot{\theta}_2) s_2 \\ \frac{1}{2} m_2 l_1 l_2 (2\dot{\theta}_1 + \dot{\theta}_2) s_2 & -\frac{1}{2} m_2 l_1 l_2 \dot{\theta}_1 s_2 \end{bmatrix}$$

$$\boldsymbol{\tau}_g(\boldsymbol{\theta}) = -g \begin{bmatrix} (m_1+m_2)l_1 s_1 + m_2 l_2 s_{12} \\ m_2 l_2 s_{12} \end{bmatrix}, \qquad \mathbf{B} = \mathbf{I}_2,$$

so both joints are directly actuated. With the example values ($m_i = l_i = 1$) these become

$$\mathbf{M} = \begin{bmatrix} 3+2c_2 & 1+c_2 \\ 1+c_2 & 1 \end{bmatrix}, \quad \boldsymbol{\tau}_g = -g \begin{bmatrix} 2s_1 + s_{12} \\ s_{12} \end{bmatrix}.$$

Second-order form. For strictly positive masses and link lengths, $\mathbf{M}(\boldsymbol{\theta})$ is positive definite and therefore invertible, so the EOM solve explicitly for the joint accelerations:

$$\ddot{\boldsymbol{\theta}} = \mathbf{M}(\boldsymbol{\theta})^{-1} \left[-\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} + \boldsymbol{\tau}_{\text{ext}} \right]. \quad (2)$$

State-space form. Stacking $\mathbf{x} = [\theta_1 \;\; \theta_2 \;\; \dot{\theta}_1 \;\; \dot{\theta}_2]^T = [\boldsymbol{\theta}^T \;\; \dot{\boldsymbol{\theta}}^T]^T$ and setting $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$, the dynamics $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ are, explicitly,

$$\mathbf{f}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \dot{\boldsymbol{\theta}} \\ \mathbf{M}(\boldsymbol{\theta})^{-1} \left[-\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} \right] \end{bmatrix}. \quad (3)$$

The measured output is the pair of joint angles, $\mathbf{y} = \mathbf{C}_y \mathbf{x}$ with $\mathbf{C}_y = [\mathbf{I}_2 \;\; \mathbf{0}_{2\times 2}]$.

B. Open-Loop Linearization

The arm hanging straight down at rest with zero torque is an equilibrium *state–input pair* $(\mathbf{x}^*, \mathbf{u}^*) = (\mathbf{0}, \mathbf{0})$: the velocities are zero and $\boldsymbol{\tau}_g(\mathbf{0}) = \mathbf{0}$, so $\mathbf{f}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$. With $\delta\mathbf{x} = \mathbf{x} - \mathbf{x}^*$ and $\delta\mathbf{u} = \mathbf{u} - \mathbf{u}^*$, the first-order Taylor approximation is

$$\boxed{\delta\dot{\mathbf{x}} = \mathbf{A}\delta\mathbf{x} + \mathbf{B}_{\text{lin}}\delta\mathbf{u},} \quad (4)$$

$$\mathbf{A} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{(\mathbf{x}^*, \mathbf{u}^*)}, \qquad \mathbf{B}_{\text{lin}} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right|_{(\mathbf{x}^*, \mathbf{u}^*)}.$$

Write the lower block of (3) as $\mathbf{M}^{-1}\mathbf{h}$ with $\mathbf{h} = -\mathbf{C}\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g + \mathbf{B}\mathbf{u}$. Then $\partial(\mathbf{M}^{-1}\mathbf{h})/\partial\boldsymbol{\theta} = (\partial\mathbf{M}^{-1}/\partial\boldsymbol{\theta})\,\mathbf{h} + \mathbf{M}^{-1}\,\partial\mathbf{h}/\partial\boldsymbol{\theta}$, and the first term vanishes because $\mathbf{h}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$. The Coriolis term $\mathbf{C}\dot{\boldsymbol{\theta}}$ is a sum of products $\dot{\theta}_i\dot{\theta}_j$ with coefficients depending on θ_2. Its derivative with respect to $\boldsymbol{\theta}$ is therefore proportional to $\dot{\theta}_i\dot{\theta}_j$, and its derivative with respect to $\dot{\boldsymbol{\theta}}$ is linear in $\dot{\boldsymbol{\theta}}$; both are zero at $\dot{\boldsymbol{\theta}} = \mathbf{0}$. Only gravity contributes, giving

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_2 \\ \mathbf{K}_g & \mathbf{0} \end{bmatrix}, \quad \mathbf{B}_{\text{lin}} = \begin{bmatrix} \mathbf{0} \\ \mathbf{M}_0^{-1}\mathbf{B} \end{bmatrix}, \quad \mathbf{K}_g = \mathbf{M}_0^{-1}\mathbf{G}, \quad (5)$$

where $\mathbf{M}_0 = \mathbf{M}(\mathbf{0})$ and $\mathbf{G} = \partial\boldsymbol{\tau}_g/\partial\boldsymbol{\theta}|_{\mathbf{0}}$, namely

$$\mathbf{M}_0 = \begin{bmatrix} (m_1+m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 & m_2 l_2 (l_1 + l_2) \\ m_2 l_2 (l_1 + l_2) & m_2 l_2^2 \end{bmatrix}, \; \mathbf{G} = -g \begin{bmatrix} (m_1+m_2)l_1 + m_2 l_2 & m_2 l_2 \\ m_2 l_2 & m_2 l_2 \end{bmatrix}.$$

Example values. For $m_i = l_i = 1$ and $g = 9.81$: $\mathbf{M}_0 = \left[\begin{smallmatrix} 5 & 2 \\ 2 & 1 \end{smallmatrix}\right]$, $\mathbf{M}_0^{-1} = \left[\begin{smallmatrix} 1 & -2 \\ -2 & 5 \end{smallmatrix}\right]$, $\mathbf{G} = -g\left[\begin{smallmatrix} 3 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, so

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -9.81 & 9.81 & 0 & 0 \\ 9.81 & -29.43 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{\text{lin}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & -2 \\ -2 & 5 \end{bmatrix}.$$

The matrix $\mathbf{G}$ is symmetric negative definite and $\mathbf{M}_0$ is positive definite, so $\mathbf{K}_g = \mathbf{M}_0^{-1}\mathbf{G}$ is similar to the symmetric negative definite matrix $\mathbf{M}_0^{-1/2}\mathbf{G}\mathbf{M}_0^{-1/2}$ and has real negative eigenvalues $-\mu_i$. Each eigenvalue λ of $\mathbf{A}$ then satisfies $\lambda^2 = -\mu_i$, i.e. $\lambda = \pm j\sqrt{\mu_i}$. For the example, $\mu = g(2 \pm \sqrt{2})$ and the eigenvalues are $\pm j2.40$ and $\pm j5.79$ rad/s, which we confirmed numerically. With no dissipation in the model, this motivates a feedback law that adds damping. We will also check controllability and observability numerically.

IV. Impedance Control

Let $\boldsymbol{\theta}_d$ be a desired (constant) joint configuration. We apply the torque law

$$\boxed{\mathbf{u} = -\boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{K}(\boldsymbol{\theta}_d - \boldsymbol{\theta}) - \mathbf{D}\dot{\boldsymbol{\theta}},} \quad (6)$$

where $\mathbf{K}, \mathbf{D} \in \mathbb{R}^{2\times 2}$ are symmetric positive definite stiffness and damping gains. Gravity is compensated, and the remaining terms act as a virtual spring-damper at each joint. Substituting into (1) gives the closed loop

$$\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \Big(\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{D}\Big)\,\dot{\boldsymbol{\theta}} + \mathbf{K}(\boldsymbol{\theta} - \boldsymbol{\theta}_d) = \boldsymbol{\tau}_{\text{ext}}.$$

With error $\mathbf{e} = \boldsymbol{\theta} - \boldsymbol{\theta}_d$ and state $\mathbf{z} = [\mathbf{e}^T \;\; \dot{\mathbf{e}}^T]^T$, linearizing about $\mathbf{z} = \mathbf{0}$, $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$ yields

$$\dot{\mathbf{z}} = \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}_d^{-1}\mathbf{K} & -\mathbf{M}_d^{-1}\mathbf{D} \end{bmatrix}}_{\mathbf{A}_{cl}} \mathbf{z} + \begin{bmatrix} \mathbf{0} \\ \mathbf{M}_d^{-1} \end{bmatrix} \boldsymbol{\tau}_{\text{ext}}, \quad (7)$$

where $\mathbf{M}_d = \mathbf{M}(\boldsymbol{\theta}_d)$. This is a mass-spring-damper whose stiffness and damping we choose. Because $\mathbf{M}_d$, $\mathbf{K}$, and $\mathbf{D}$ are positive definite, $\mathbf{A}_{cl}$ is Hurwitz. For the example arm with $\boldsymbol{\theta}_d = \mathbf{0}$, $\mathbf{K} = 20\,\mathbf{I}$ and $\mathbf{D} = 5\,\mathbf{I}$, the eigenvalues of $\mathbf{A}_{cl}$ are about -24.4, -4.79, and $-0.43 \pm j1.80$. We will verify this in simulation. To keep the project tractable, we use diagonal gains and simulate in joint space only.

Planned analysis and simulation

- Eigenvalues of $\mathbf{A}_{cl}$ for a few choices of $\mathbf{K}$ and $\mathbf{D}$ at one or two values of $\boldsymbol{\theta}_d$.
- Nonlinear simulation (numerical ODE integration): step in $\boldsymbol{\theta}_d$ and a short disturbance torque pulse $\boldsymbol{\tau}_{\text{ext}}$, compared against the linear model.
- Effect of stiffness versus damping on the response.

V. Plan & Deliverables

We will implement the nonlinear model and controllers in Python (NumPy/SciPy) and report the open-loop and closed-loop eigenvalues and the step and disturbance responses for a few gain choices. Success means a stable, well-damped closed loop that tracks $\boldsymbol{\theta}_d$ and deflects predictably under $\boldsymbol{\tau}_{\text{ext}}$. All work is in simulation, with no hardware.

TABLE II

PROPOSED TIMELINE.

Stage	Task
1	Model, linearize, and verify numerically
2	Open-loop eigenvalue, controllability, observability
3	Impedance controller, closed-loop analysis, simulation
4	Final report and presentation

REFERENCES

[1] N. Hogan, "Impedance control: An approach to manipulation: Part i—theory," *Journal of Dynamic Systems, Measurement, and Control*, vol. 107, no. 1, pp. 1–7, 03 1985. [Online]. Available: https://doi.org/10.1115/1.3140702

[2] R. Tedrake, *Underactuated Robotics*, 2023. [Online]. Available: https://underactuated.csail.mit.edu